5	Neural Networks	Lecture	Exam, Assignment
6	Recurrent Neural Networks	Lecture	Exam, Assignment
7			
8	Seq2seq models and Attention	Lecture	Exam, Project
9	Transformers	Lecture	Exam, Project
10	Large Language Models	Lecture	Exam, Project
11	Generative LLMs	Lecture	Exam, Project
12	Generative LLMs	Lecture	Exam, Project
13	Ethical NLP	Lecture	Exam

CSE752: Deep Neural Networks

Course Catalog Description (Content):

This course provides an in-depth exploration of deep learning. Topics include neural network basics, activation functions, forward and backward propagation; architectures like multi-layer perceptron (MLP) and fully connected layers; convolutional neural networks (CNNs) for vision tasks; recurrent neural networks (RNNs) including LSTMs and GRUs for sequential data; unsupervised learning methods like autoencoders, variational autoencoders (VAEs), and generative adversarial networks (GANs); transformer models such as BERT and GPT; optimization techniques including gradient descent variants, stochastic gradient descent (SGD), Adam, and learning rate schedules; regularization techniques like dropout and batch normalization; advanced architectures including ResNets and capsule networks; and transfer learning strategies for fine-tuning pre-trained models.

Rationale of the Course:

Deep learning is a rapidly evolving field with significant applications in various domains such as computer vision, natural language processing, and generative AI. This course is designed to equip students with the theoretical understanding and practical skills necessary to develop and implement deep neural network models for solving real-world problems.

Course Objective:

The core objectives of this course are to:

- To introduce fundamental concepts and architectures of deep learning.
- To implement and train deep learning models using modern frameworks.
- To provide an understanding of advanced architectures and optimization techniques.

- To explore applications of deep learning in different domains.
- To analyze the challenges and ethical considerations of deep learning models.

Class Week	Topic	Teaching-Learning Strategy	Assessment Strategy
1	Introduction to Deep Learning	Lecture, Discussion	Quiz, Mid, Assignment
2	Neural Networks: Basics & Activation Functions, Loss Functions	Lecture, Discussion, Hands-on Exercise	Quiz, Mid, Assignment
3	Forward & Backward Propagation	Lecture, Hands-on Exercise	Assignment, Quiz, Mid
4	Multi-Layer Perceptron (MLP)	Lecture,Discussion	Quiz , Mid, Project Milestone
5	Convolutional Neural Networks (CNNs),	Lecture, Case Studies	Final,Assig nment
6	Pooling, Filters, Applications in Vision	Lecture, Hands-on Exercise	Final,Assig nment
7	Recurrent Neural Networks (RNNs)	Lecture, Hands-on Practice	Quiz,Mid
8	Midterm Examination		
9	Long Short-Term Memory (LSTM) & GRUs, Autoencoders and Variational Autoencoders	Lecture	Assignment, M Quiz, Final
10	Generative Adversarial Networks (GANs), Transformers, Self-Attention, BERT, GPT	Lecture	Assignment, Quiz, Final
11	Optimization Techniques & Regularization	Lecture	Project, Final
12	Advanced Architectures (ResNets, Capsule Networks)	Lecture	Final, Assignment

13	Transfer Learning & Domain Adaptation	Project , Lecture , Discussion	Project Presentation
14	Final Examination		

CSE753: Advanced Reinforcement Learning

Course Catalog Description (Content):

Fundamentals of Reinforcement Learning: Markov decision processes (MDPs), reward structures, exploration vs. exploitation; Value-Based Methods: Q-learning, deep Q-networks (DQNs), double DQN, dueling DQN; Policy-Based Methods: policy gradient methods, REINFORCE algorithm, actor-critic; Model-Based Reinforcement Learning: planning, world models, model-predictive control (MPC); Multi-Agent Reinforcement Learning: cooperative and competitive agents, multi-agent learning frameworks; Deep Reinforcement Learning: deep Q-learning, trust region policy optimization (TRPO), proximal policy optimization (PPO); Hierarchical Reinforcement Learning: options framework, hierarchical DQN, learning sub-policies; Exploration Strategies: epsilon-greedy, Thompson sampling, curiosity-driven exploration; Inverse Reinforcement Learning: inverse optimization, learning from demonstration; Safe and Robust Reinforcement Learning: reward shaping, safe exploration, robust control; Imitation Learning: behavior cloning, generative adversarial imitation learning (GAIL); Applications: robotics, autonomous vehicles, game playing, recommendation systems; Ethical Considerations: fairness, reward specification, interpretability in RL.

Rationale of the Course:

Advanced Reinforcement Learning (RL) is at the forefront of AI research, enabling agents to learn optimal behaviors through interaction with complex environments. This course delves into cutting-edge RL techniques, covering value-based, policy-based, and model-based methods, along with multi-agent, hierarchical, and safe RL. Students will explore deep RL algorithms, imitation learning, and ethical considerations, gaining the expertise to design and implement sophisticated RL systems for applications in robotics, autonomous vehicles, and beyond.

Course Objective:

The core objectives of this course are to:

- To provide in-depth knowledge of advanced RL algorithms and techniques, including deep RL and multi-agent RL.\
- To develop proficiency in implementing and applying Q-learning, DQN, policy gradient, and actor-critic methods.
- To understand and implement planning algorithms and world models for efficient RL.
- To gain expertise in designing and implementing RL systems for cooperative and competitive agents, and hierarchical tasks.
- To apply deep RL algorithms like TRPO and PPO to solve complex control problems.
- To explore methods for learning from demonstrations and inferring reward functions.